

AIM:

The primary aim of using plant polysaccharides in cancer treatment is to develop effective, safe, and natural therapeutic agents that can enhance the efficacy of conventional cancer therapies, reduce side effects, and improve patient outcomes. Plant polysaccharides are explored for their potential to modulate the immune system, inhibit tumor growth, and provide a complementary approach to existing cancer treatments.

OBJECTIVES:

- 1. Immunomodulation:** To investigate the ability of plant polysaccharides to modulate the immune system, enhancing the body's natural defense mechanisms against cancer cells. This includes stimulating the activity of immune cells such as macrophages, natural killer (NK) cells, and T-cells.
- 2. Anti-Tumor Activity:** To evaluate the direct anti-tumor effects of plant polysaccharides, including their ability to induce apoptosis (programmed cell death), arrest the cell cycle, and inhibit angiogenesis (formation of new blood vessels that supply nutrients to tumors).
- 3. Combination Therapy:** To explore the synergistic effects of combining plant polysaccharides with conventional chemotherapeutic drugs. This objective aims to enhance the efficacy of chemotherapy, reduce drug dosage, and minimize side effects.
- 4. Safety and Efficacy:** To assess the safety and efficacy of plant polysaccharides in preclinical and clinical studies. This includes evaluating their pharmacokinetics, bioavailability, and potential toxicities.
- 5. Mechanistic Studies:** To elucidate the molecular mechanisms underlying the anti-cancer effects of plant polysaccharides. This involves studying their interactions with cellular pathways, signaling molecules, and the tumor microenvironment.
- 6. Development of Novel Therapeutics:** To develop novel cancer therapeutics based on plant polysaccharides. This includes formulating polysaccharide-based drugs, optimizing their delivery systems, and conducting clinical trials to validate their therapeutic potential.

HISTORY OF HERBAL DRUGS IN CANCER TREATMENT:

Ancient and Traditional Use :

1. Traditional Chinese Medicine (TCM): Herbal remedies have been used in TCM for thousands of years to treat various ailments, including cancer. Herbs like *Artemisia annua* (sweet wormwood) and *Camptotheca acuminata* (happy tree) have been traditionally used for their anti-cancer properties.

2. Ayurveda: In India, Ayurveda has utilized herbs such as *Withania somnifera* (ashwagandha) and *Curcuma longa* (turmeric) for their potential anti-cancer effects.

3. Ancient Egypt and Greece: Ancient Egyptians and Greeks used herbs like *Aloe vera* and *Glycyrrhiza glabra* (licorice) for their medicinal properties, including cancer treatment.[5]

Modern Developments :

1. Paclitaxel (Taxol): Derived from the bark of the Pacific yew tree (*Taxus brevifolia*), paclitaxel is one of the most successful plant-based anti-cancer drugs. It was discovered in the 1960s and has been widely used in chemotherapy.

2. Vincristine and Vinblastine: These alkaloids, derived from the *Catharanthus roseus* (Madagascar periwinkle), have been used in the treatment of various cancers, including leukemia and lymphoma.

3. Camptothecin: Extracted from the *Camptotheca acuminata* tree, camptothecin and its derivatives (e.g., irinotecan and topotecan) are used in the treatment of colorectal and ovarian cancers.[6]

Current Research and Future Directions :

1. Natural Product-Based Drug Discovery: Research continues to explore the potential of natural products in cancer therapy. Advances in technology and pharmacology have led to the discovery of new bioactive compounds from plants, marine organisms, and microorganisms.

2. Combination Therapies: Herbal drugs are being investigated for their potential to enhance the efficacy of conventional cancer treatments and reduce side effects. For example,

combining curcumin with chemotherapy has shown promising results in improving therapeutic outcomes.[7]

Challenges and Considerations :

- 1. Standardization and Quality Control:** Variability in the composition and quality of herbal products can affect their efficacy and safety. Standardization of extraction and purification methods is essential.
- 2. Clinical Trials:** Rigorous clinical trials are needed to validate the safety and efficacy of herbal drugs in cancer treatment. This will help in gaining regulatory approval and acceptance in mainstream oncology practice.[6]

CANCER A LEADING THREAT :

Introduction:

Cancer is a group of diseases characterized by the uncontrolled growth and spread of abnormal cells. If the spread is not controlled, it can result in death. Cancer can affect almost any part of the body and has many different forms, each with its own unique characteristics and treatment options.[2]

Types of Cancer:

- 1. Carcinomas:** These cancers originate in the epithelial cells that line the internal organs and skin. Examples include breast, lung, and colon cancer.
- 2. Sarcomas:** These cancers arise from connective tissues such as bone, muscle, and fat. Examples include osteosarcoma and liposarcoma.
- 3. Leukemias:** These cancers start in the blood-forming tissues, such as the bone marrow, and result in the production of large numbers of abnormal blood cells.
- 4. Lymphomas:** These cancers originate in the lymphatic system, which is part of the body's immune system. Examples include Hodgkin lymphoma and non-Hodgkin lymphoma.
- 5. Central Nervous System Cancers:** These cancers begin in the tissues of the brain and spinal cord.[1]



Fig: Squamous cell carcinoma occluding bronchus, with lymph node metastasis.

https://upload.wikimedia.org/wikipedia/commons/thumb/e/e5/Lymph_node_metastasis%2C_squamous_cell_carcinoma_bromchus.jpg/800px-Lymph_node_metastasis%2C_squamous_cell_carcinoma_bromchus.jpg

Causes and Risk Factors:

Cancer can be caused by a variety of factors, including genetic mutations, environmental exposures, and lifestyle choices. Some common risk factors include:

- **Genetic Predisposition:** Certain inherited genetic mutations can increase the risk of developing cancer.

- **Environmental Exposures:** Exposure to carcinogens such as tobacco smoke, asbestos, and ultraviolet radiation can increase cancer risk.
- **Lifestyle Factors:** Poor diet, lack of physical activity, and excessive alcohol consumption can contribute to cancer development.
- **Infections:** Certain infections, such as human papillomavirus (HPV) and hepatitis B and C, can increase the risk of specific cancers.[2]

Symptoms:

The symptoms of cancer vary depending on the type and location of the cancer. Common symptoms include:

- Unexplained weight loss
- Fatigue
- Pain
- Skin changes
- Changes in bowel or bladder habits
- Persistent cough or trouble breathing
- Unusual bleeding or discharge[2]

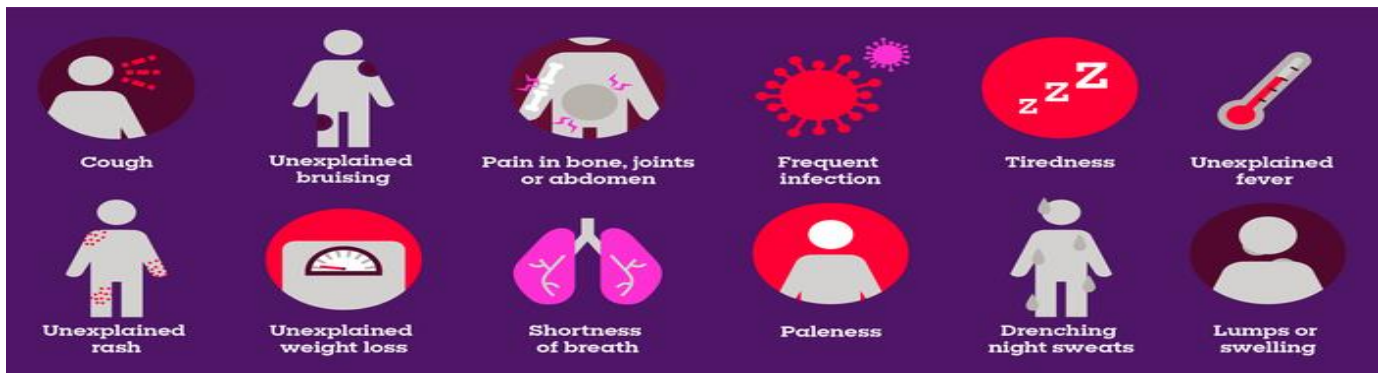


Fig : Symptoms of Cancer

https://bcuk.cdn.ngo/images/blood_cancer_symptoms.width-700.png

Diagnosis:

Cancer diagnosis typically involves a combination of methods, including:

- **Imaging Tests:** X-rays, CT scans, MRI, and PET scans are used to visualize tumors.

- **Biopsy:** A sample of tissue is taken from the suspected tumor and examined under a microscope to confirm the presence of cancer cells.
- **Laboratory Tests:** Blood, urine, and other tests can help detect cancer markers and assess overall health.[3]

Treatment:

Cancer treatment depends on the type, stage, and location of the cancer, as well as the patient's overall health. Common treatment options include:

- **Surgery:** Removal of the tumor and surrounding tissue.
- **Radiation Therapy:** Use of high-energy radiation to kill cancer cells.
- **Chemotherapy:** Use of drugs to kill cancer cells or stop their growth.
- **Immunotherapy:** Boosting the body's immune system to fight cancer.
- **Targeted Therapy:** Drugs that target specific molecules involved in cancer growth and spread.
- **Hormone Therapy:** Blocking hormones that fuel certain cancers, such as breast and prostate cancer.[3]

Prevention:

While not all cancers can be prevented, certain measures can reduce the risk:

- Avoiding tobacco use- Maintaining a healthy diet and weight
- Staying physically active
- Limiting alcohol consumption
- Protecting skin from excessive sun exposure
- Getting vaccinated against cancer-related infections (e.g., HPV, hepatitis B).[4]



Fig: Prevention of Cancer

https://www.grhosp.on.ca/assets/images/general/_contentLarge/CancerPrevention.jpg

PLANT POLYSACCHARIDES:

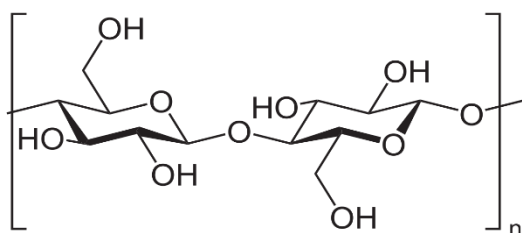
Plant Polysaccharides are complex carbohydrates composed of long chains of monosaccharide units linked by glycosidic bonds. They are abundant in nature and can be found in plants, animals, bacteria and fungi. Among them, Plant Polysaccharides have garnered significant attention due to their diverse biological activities and potential applications in various fields, including medicine, food and cosmetics. This paper provides an overview of Plant Polysaccharides, focusing on their types, biological activities and applications.

TYPES:

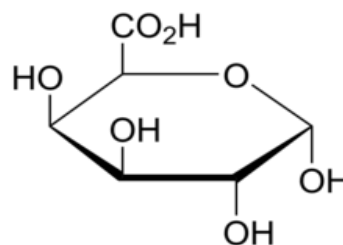
Plant Polysaccharides can be broadly classified in to several types based on their structure and function :

- 1. Cellulose:** The most abundant Polysaccharides in nature, cellulose provides structural support to plant cell walls.[10]
- 2. HemiCellulose:** A heterogenous group of Polysaccharides that, along with cellulose, contributes to the structural integrity of plant cell walls.[10]
- 3. Pectin:** A Polysaccharide rich in galacturonic acid, Pectin is found in the Primary cell walls of Plants and is widely used in the food industry as a gelling agent.[16]
- 4. Starch:** A storage Polysaccharide composed of amylose and amylopectin, starch serves as an energy reserve in Plants.[8]
- 5. Gums and Mucilages:** These Polysaccharides are secreted by plants and have various applications in the food and Pharmaceutical industries.[14]

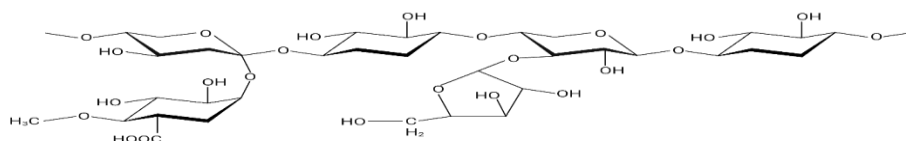
Cellulose Chemical Structure



Pectin Chemical Structure



Hemicellulose Chemical Structure



BIOLOGICAL ACTIVITY:

Plant Polysaccharides exhibit a wide range of biological activities including:

- **Antioxidant:** Polysaccharides can scavenge free radicals and protect cells from oxidative damage.[11]
- **Anti-Inflammatory:** They can modulate the immune response and reduce inflammation.[18]
- **Anti tumor:** Some Polysaccharides have shown Potential in inhibiting tumor growth and enhancing the immune system's ability to fight cancer.[17]
- **Antimicrobial:** Polysaccharides can inhibit the growth of various pathogens, including bacteria and fungi.[19]
- **Antidiabetic:** They can help regulate blood sugar levels and improve insulin sensitivity.[13]

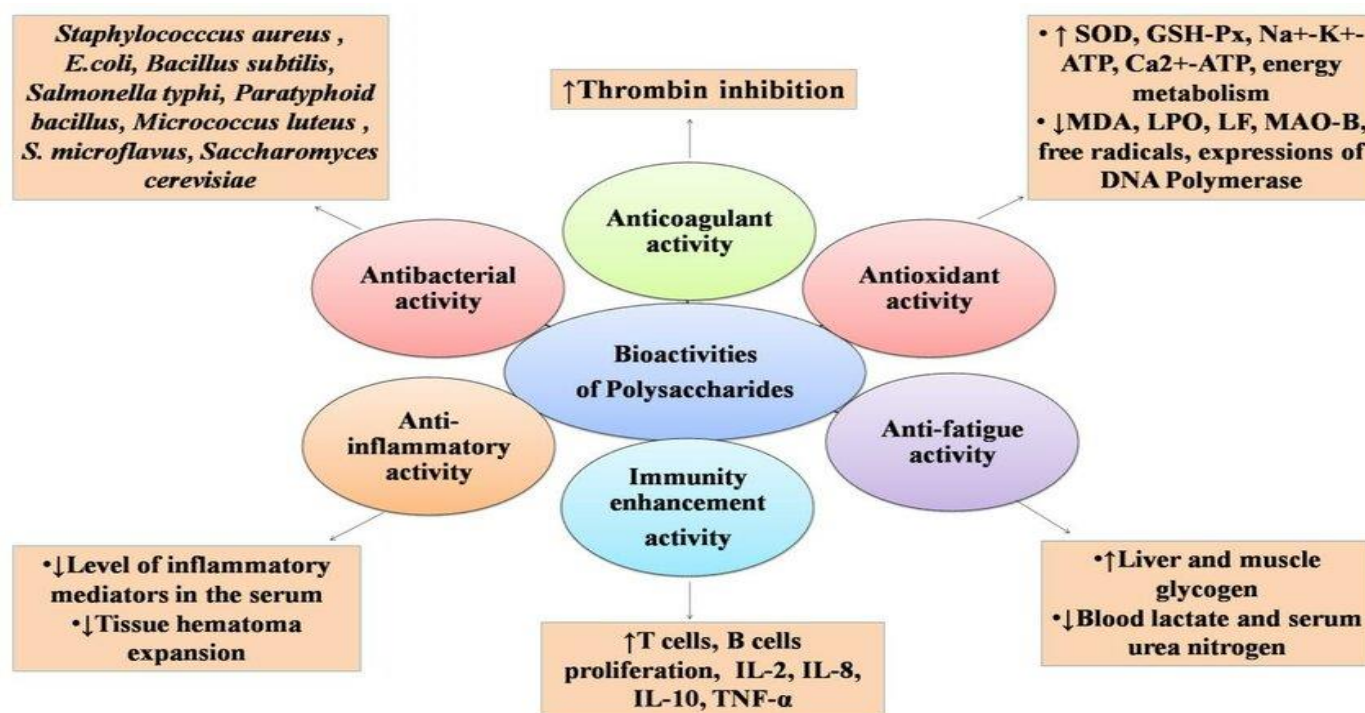


Figure : Biological activities of Plant Polysaccharides

<https://www.researchgate.net/publication/355382296/figure/fig3/AS:11431281102467438@1669467264472/Bioactivities-of-polysaccharides-and-their-biological-actions.jpg>

PLANT POLYSACCHARIDES IN THE USE OF CANCER TREATMENT:

ORIGIN:

Plant Polysaccharides are naturally occurring complex carbohydrates found in various plants. They are extracted from different parts of plants, including roots, stems, leaves and fruits. Some common sources of Plant Polysaccharides include mushrooms, algae and traditional medicinal plants.

MECHANISMS OF ACTION:

It act by various mode of action that acts upon the cells in order to act as:

- 1. Anti-tumor activity:** Plant Polysaccharides exhibit significant anti-tumor effects by enhancing immunoregulation, inhibiting tumor cell growth and preventing tumor cell invasion and metastasis. They achieve this through mechanisms such as cell cycle arrest, anti-angiogenesis, apoptosis and immunomodulation.[20,21]
- 2. Immunomodulation:** Polysaccharides can modulate the immune system, particularly by targeting macrophage polarization. This involves promoting the M1 Phenotype, which suppresses cancer growth and inhibiting the M2 Phenotype, which supports tumor Progression.[21]
- 3. Gut microbiota Interaction:** The interaction between dietary Polysaccharides and gut microbiota plays a crucial role in cancer prevention. Polysaccharides can influence the gut microbiota composition, leading to enhanced immune responses and reduce cancer risk.[22]

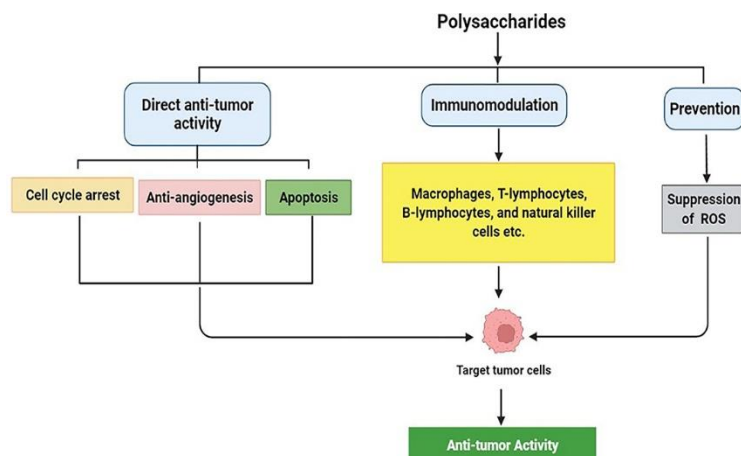


Figure: Polysaccharides Mechanisms

https://www.frontiersin.org/files/Articles/838179/fnut-09-838179-HTML-r1/image_m/fnut-09-838179-g001.jpg

RECENT DEVELOPMENTS IN CANCER TREATMENT USING PLANT POLYSACCHARIDES:

- 1. Synergistic Effects with Chemotherapy:** Recent studies have explored the combination of Plant Polysaccharides with conventional chemotherapeutic drugs. This combination has shown to enhance the efficacy of chemotherapy while reducing its side effects. For Example, a review highlighted the Potential of Plant Polysaccharides to improve chemotherapeutic outcomes in breast cancer.[20]
- 2. Novel Dosage Form:** Researchers have been developing new dosage forms of Plant Polysaccharides to improve their bioavailability and therapeutic efficacy. These novel formulations include nanoparticles, liposomes and hydrogels which can enhance the delivery of Polysaccharides to tumor sites and improve their anti cancer Effects.[23]
- 3. Immunomodulatory Properties:** Plant Polysaccharides have been found to modulate the immune system, particularly by targeting macrophage polarization. This involves promoting the M1 Phenotype, which suppresses cancer growth and inhibiting the M2 Phenotype, which supports tumor Progression.[24]
- 4. Antioxidant and Anti-Inflammatory Activities:** The antioxidant and anti-inflammatory properties of Plant Polysaccharides contribute to their potential as therapeutic agents in cancer treatment. These activities help in reducing oxidative stress and inflammation, which are associated with cancer Progression.[24]

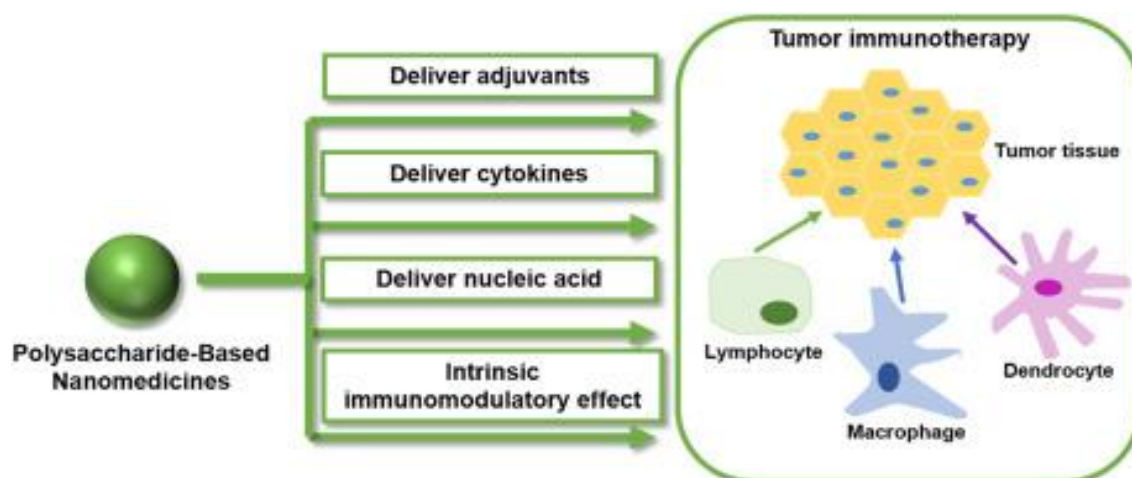


Figure : Nano medicines based upon Plant Polysaccharides

<https://ars.els-cdn.com/content/image/1-s2.0-S2452199X21001092-ga1.jpg>

ACTUAL USE OF PLANT POLYSACCHARIDES IN CANCER TREATMENT:

Plant Polysaccharides have shown promising potential in cancer treatment due to their bioactive properties. These natural compounds can enhance the effectiveness of conventional chemotherapy while reducing in side effects. Here are some key points about their actual use:

- **Synergistic Effect:**

Plant Polysaccharides can work synergistically with chemotherapeutic drugs, increasing their cytotoxicity to cancer cells and reducing drug dose.[20]

- **Mechanism of Action:**

They exhibit anti-tumor activities through various mechanisms, including cell cycle arrest, anti-angiogenesis, apoptosis and immunomodulation.[25]

- **Preclinical Studies:**

Research has highlighted the potential of Plant Polysaccharides in improving chemotherapeutic outcomes in Breast cancer, paving the way for more effective and safer treatment strategies.[20]

- **Non-Toxic Characteristics:**

These bioactive macromolecules are considered potential candidates against cancer due to their non toxic characteristics.[25]

IDEAL PLANT POLYSACCHARIDES:

Plant Polysaccharides have shown significant Potential in cancer treatment due to their bioactive properties. Here are some ideal Plant Polysaccharides used in Cancer Therapy:

- 1. β – Glucans:** Found in mushrooms, oats and barley. These have immunomodulatory and anti-tumor properties. They enhance the immune system's ability to fight cancer cells and improve the effectiveness of chemotherapy.

Advantages:

- Enhances the immune system's ability to fight cancer cells.
- Improves the effectiveness of chemotherapy.
- Non-Toxic and safe for consumption.

Disadvantages:

- Limited bioavailability when taken orally.
- Requires further research to fully understand its mechanisms.[20]

2. Pectins: Derived from fruits like apples and citrus, pectins have been shown to inhibit tumor growth and metastasis. They also possess antioxidant properties that protect normal cells from damage during chemotherapy.

Advantages:

- Inhibits tumor growth and metastasis.
- Possesses antioxidant properties that protect normal cells.
- Can be easily extracted from fruits like apples and citrus.

Disadvantages:

- Variability in composition depending on the sources.
- Requires high doses for significant effects.[25]

3. Arabinogalactans: Found in plants like Echinacea and larch trees, arabinogalactans have immunomodulatory effects and can enhance the body's natural defense mechanisms against cancer.

Advantages:

- Enhances the body's natural defense mechanisms against cancer.
- Immunostimulatory effects.
- Non-toxic and safe for consumption.

Disadvantages:

- Limited clinical studies on its effectiveness.
- Requires further research to determine optimal dosages.[20]

4. Fucodians: Extracted from brown seaweed, fucodians exhibit anti-tumor, anti angiogenic and immunomodulatory activities. They can inhibit cancer cell proliferation and induce apoptosis.

Advantages:

- Exhibits anti-tumor, anti angiogenic and immunomodulatory activities.
- Inhibits cancer cell proliferation and induces apoptosis.
- Extracted from brown seaweed, a renewable resource.

Disadvantages:

- Variability in composition, depending on the source.
- Requires further research to fully understood.[25]

5. Inulin: Found in Chicory root and Jerusalem artichoke, inulin has prebiotic properties that support gut health and enhance the immune system. It also exhibits anti-tumor effects.

Advantages:

- Supports gut health and enhances the immune system.
- Exhibits anti-tumor effects.

Disadvantages:

- Limited bioavailability when taken orally.
- Requires further research to determine optimal dosage.[25]

WHY WE USE PLANT POLYSACCHARIDES?

Plant Polysaccharides are increasing being recognized for their potential in cancer treatment due to their various bioactive properties. Here are some reasons why they are used:

1. Anti-tumor activity: Polysaccharides have been shown to exhibit anti-tumor activities. They can inhibit the growth of cancer cells and induce apoptosis(program cell death) in the cells.[21]

2. Immunomodulation: These compounds can modulate the immune system, enhancing the body's natural ability to fight cancer. They can stimulate the activity of immune cells such as macrophages, natural killer cells and T-cells.[25]

3. Non-toxic characteristics: Unlike many conventional cancer treatments, Plant Polysaccharides are generally non-toxic and have fewer side effects. This makes them a safer alternative or complementary option in cancer therapy.[21]

4. Anti-angiogenesis: Polysaccharides can inhibit the formation of new blood vessels that tumors need to grow and spread, a process known as angiogenesis.[25]

CONCLUSION:

Plant Polysaccharides have emerged as promising candidates in the development of novel cancer therapies. Their diverse biological activities, including immunomodulatory, anti-tumor, antioxidant and anti-inflammatory effects, make them valuable resources for cancer treatment. These natural compounds can enhance the immune response,

regulate the tumor microenvironment and inhibit tumor growth through various mechanisms, such as cell cycle arrest, apoptosis and anti-angiogenesis.

One of the most significant advantages of the plant polysaccharides is their ability to modulate the immune system. They stimulate the activity of various immune cells, including macrophages, natural killer cells, and T-cells, enhancing the body's ability to recognize and attack cancer cells. This immunomodulatory effect is particularly beneficial in cancer therapy, as it helps to boost the patient's immune response against tumors.

Additionally, plant polysaccharides have demonstrated direct anti tumor activity by including apoptosis(programmed cell death) and arresting the cell cycle. These mechanisms help to prevent the proliferation and spread of cancer cells, effectively inhibiting tumor growth. Furthermore, their anti-angiogenic properties inhibit the formation of new blood vessels that supply nutrients to tumors, thereby starving and shrinking them.

The combination of plant polysaccharides with the conventional chemotherapeutic drugs has shown synergistic effects, improving therapeutic outcomes and reducing side effects. This approach offers a promising strategy for developing safer and more effective cancer treatments. By targeting multiple pathways involved in cancer progression, plant polysaccharides enhance the overall efficacy of cancer therapy.

However, there are challenges to the use of plant polysaccharides in cancer treatment, including standardization, bioavailability and the need for rigorous clinical trials. Standardization of extraction and purification methods is essential to ensure consistent composition and activity. Improving the bioavailability and stability of polysaccharides is crucial for their effective use in cancer therapy. Advanced delivery systems, such as nanoparticles, are being developed to address these issues. More rigorous clinical trials are needed to validate the safety and efficacy of plant polysaccharides in cancer treatment, facilitating their regulatory approval and acceptance in mainstream oncology practice.

In conclusion, plant polysaccharides represent a valuable and promising resource in the development of novel cancer therapies. Their diverse biological activities, combined with the potential for synergistic effects with conventional treatments, make them promising candidates for improving patient outcomes and reducing adverse effects. Continued

research and development to this field hold great potential for advancing cancer therapy and enhancing the quality of life for cancer patients.

REFERENCE:

01. 2006 current medical diagnosis and treatment- 45th edition by Lawrence M. Tiereney. Jr
02. American Association for Cancer Research <https://aacrjournals.org/cancerres>
03. Journal of Cancer <https://www.icancer.org/>
04. BMC Cancer <https://bmccancer.biomedcentral.com/articles/10.1186/s12885-024-13035-6>
05. Natural Products as Treatment against Cancer: A Historical and Current Vision <https://www.clinicsinoncology.com/open-access/natural-products-as-treatment-against-cancer-a-historical-and-current-vision-1716.pdf>
06. Natural Products in Cancer Therapy: Past, Present and Future <https://link.springer.com/article/10.1007/s13659-020-00293-7>
07. HERBAL TREATMENT OF CANCER <https://ijcrt.org/papers/IJCRT23A5108.pdf>
08. Bai, L., Zhu, P., Wang, W., & Wang, M. (2020). The influence of extraction pH on the chemical compositions, macromolecular characteristics, and rheological properties of polysaccharide: The case of okra polysaccharide. Food Hydrocolloids, 102, 105586.
09. Cai, C., Wang, Y., Yu, W., Wang, C., Li, F., & Tan, Z. (2020). Temperature-responsive deep eutectic solvents as green and recyclable media for the efficient extraction of polysaccharides from Ganoderma lucidum. Journal of Cleaner Production, 274, 123047.
10. Caffall, K. H., & Mohnen, D. (2009). The structure, function, and biosynthesis of plant cell wall pectic polysaccharides. Carbohydrate Research, 344(14), 1879-1900.
11. Chen, C., Wang, P. P., Huang, Q., You, L. J., Liu, R. H., Zhao, M. M., Fu, X., & Luo, Z. G. (2019). A comparison study on polysaccharides extracted from: Fructus Mori using different methods: Structural characterization and glucose entrapment. Food and Function, 10(6), 3684-3695.
12. Chen, R., Liu, Z., Zhao, J., Chen, R., Meng, F., Zhang, M., & Ge, W. (2011). Antioxidant and immunobiological activity of water-soluble polysaccharide fractions purified from Acanthopanax senticosus. Food Chemistry, 127(2), 434-440.
13. Eseyin, O. (2014). Hypoglycemic Potential of Polysaccharides of the Leaf Extract of Telfairia occidentalis. Annual Research & Review in Biology, 4(11), 1813-1826.

14. Gao, Q. H., Fu, X., Zhang, R., Wang, Z., & Guo, M. (2018). Neuroprotective effects of plant polysaccharides: A review of the mechanisms. *International Journal of Biological Macromolecules*, 106, 749-754.
15. Gong, G., Dang, T., Deng, Y., Han, J., Zou, Z., Jing, S., & Wang, Z. (2018). Physicochemical properties and biological activities of polysaccharides from *Lycium barbarum* prepared by fractional precipitation. *International Journal of Biological Macromolecules*, 109, 611-618.
16. Jia, X., Ma, L., Li, P., Chen, M., & He, C. (2016). Prospects of *Poria cocos* polysaccharides: Isolation process, structural features and bioactivities. *Trends in Food Science and Technology*, 54, 52-62.
17. Jin, M., Huang, Q., Zhao, K., & Shang, P. (2013). Biological activities and potential health benefit effects of polysaccharides isolated from *Lycium barbarum* L. *International Journal of Biological Macromolecules*, 54(1), 16-23.
18. Lee, E., Yun, N., Jang, Y. P., & Kim, J. (2013). *Lilium lancifolium* Thunb. extract attenuates pulmonary inflammation and air space enlargement in a cigarette smoke-exposed mouse model. *Journal of Ethnopharmacology*, 149(1), 148-156.
19. Wu, W., Zhu, Y., Zhang, L., Yang, R., & Zhou, Y. (2012). Extraction, preliminary structural characterization, and antioxidant activities of polysaccharides from *Salvia miltiorrhiza* Bunge. *Carbohydrate Polymers*, 87(2), 1348-1353.
20. The Potential of Plant Polysaccharides and Chemotherapeutic Drug Combinations in the Suppression of Breast Cancer <https://www.mdpi.com/1422-0067/25/22/12202>
21. Natural plant derived polysaccharides targeting macrophage polarization <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2024.1408377/full>
22. <https://www.mdpi.com/2072-6643/15/13/3019>
23. Application of natural polysaccharides and their novel dosage form over gynecological cancer <https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2023.1195104/full>
24. <https://link.springer.com/article/10.1007/s11101-024-09940-0>
25. Polysaccharides as Potential Antitumor Biomacromolecules Review <https://www.frontiersin.org/journals/nutrition/articles/10.3389/fnut.2022.838179/full>
26. Banani Kundu, Rui L. Reis, Subhas C. Kundu. (2022). Polysaccharides in Cancer Therapy. In: Oliveira, J.M., Radhouani, H., Reis, R.L. (eds) *Polysaccharides of Microbial Origin*. Springer, Cham. https://link.springer.com/referenceworkentry/10.1007/978-3-030-42215-8_42

27. Yin M, Zhang Y, Li H. Advances in research on immunoregulation of macrophages by plant polysaccharides. *Front Immunol.* 2019;10:145.
<https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2024.1408377/full>
28. Khan T, Date A, Chawda H, Patel K. Polysaccharides as potential anticancer agents—a review of their progress. *Carbohydr Polym.* 2019;210:412–428.
<https://www.mdpi.com/1422-0067/25/22/12202>